

Digital Twin and IoT for Smart Campus/Infrastructure

Focus Area: Architecture

R1: Intelligent Campus System Design Based on Digital Twin (2022)

Problem: Traditional campus systems are fragmented and lack integrated monitoring.

Why this approach: Five-layer architecture consisting of Physical Domain, Virtual Domain, Data Platform, Service Platform, and Connection Layer.

Result: Enables real-time synchronization and smart campus management.

Review: A digital twin architecture that integrates IoT devices, data platforms, and campus services into a unified system.

Link: <https://www.mdpi.com/2079-9292/11/21/3437>

R2: A Dynamic Digital Twin Framework for Sustainable Facility Management in a Smart Campus (2025)

Problem: Facility information is distributed across multiple systems.

Why this approach: Combines BIM, GIS, IoT, and Digital Twin technologies.

Result: Improves facility monitoring, maintenance, and sustainability management.

Review: Demonstrates a BIM-GIS-IoT architecture suitable for smart campus operations.

Link: <https://www.mdpi.com/2227-7080/13/10/439>

R3: A Smart Campus' Digital Twin for Sustainable Comfort Monitoring (2020)

Problem: Monitoring comfort conditions across campus buildings is difficult.

Why this approach: Uses IoT sensors connected to a campus digital twin.

Result: Provides real-time environmental monitoring.

Review: Shows how IoT and digital twins can support campus-wide monitoring.

Link: <https://www.mdpi.com/2071-1050/12/21/9196>

R4: A Framework for Scalable Digital Twin Deployment in Smart Campus Building Facility Management

Problem: Large-scale deployment of digital twins is challenging.

Why this approach: Introduces a scalable facility-management architecture.

Result: Supports deployment across multiple campus buildings.

Review: Useful reference for digital twin architecture scalability.

Link: <https://arxiv.org/abs/2512.12149>

R5: A Web of Things Architecture for Digital Twin Creation and Model-Based Reinforcement Control

Problem: Interoperability between devices and platforms is limited.

Why this approach: Uses Web of Things standards for integration.

Result: Improves communication and interoperability.

Review: Demonstrates standards-based digital twin architecture.

Link: <https://arxiv.org/abs/2301.12761>

R6: A Comprehensive Review of Digital Twin – Part 1

Problem: Lack of a common understanding of digital twin architectures.

Why this approach: Reviews digital twin frameworks and enabling technologies.

Result: Identifies common architecture patterns.

Review: Useful foundation for architecture classification.

Link: <https://arxiv.org/abs/2208.14197>

R7: Digital Twin: Enabling Technologies, Challenges and Open Research

Problem: Digital twin adoption lacks standardization.

Why this approach: Examines technologies and architecture models.

Result: Summarizes challenges and research opportunities.

Review: Provides architecture concepts applicable to smart campuses.

Link: <https://arxiv.org/abs/1911.01276>

R8: Developing Campus Digital Twin Using Interactive Visual Analytics Approach

Problem: Campus data visualization and integration remain difficult.

Why this approach: Combines analytics, visualization, and digital twin concepts.

Result: Improves decision support and monitoring.

Review: Useful example of architecture supporting campus analytics.

Link: <https://link.springer.com/article/10.1007/s44243-024-00033-2>

R9: Application of Digital Twin in Campus: Synergistic Effects of IoT, BIM, and GIS

Problem: Campus information exists in isolated systems.

Why this approach: Integrates IoT, BIM, GIS, and DT technologies.

Result: Creates a unified digital platform.

Review: Shows how multiple technologies are combined in campus architecture.

Link: <https://journals.sagepub.com/doi/10.3233/ATDE250277>

R10: The Development of a 3D Model Based on Digital Twin for Campus Master Plan Optimization

Problem: Campus planning requires integrated digital representations.

Why this approach: Uses digital twin architecture and 3D campus models.

Result: Supports planning and optimization.

Review: Demonstrates architecture for long-term campus management.

Link: <https://link.springer.com/10.1007/s44327-026-00236-w>